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Phase 3: New Features

This phase introduced two examples of new features that can be added to the current Trojan Book. Example 1 enlarges the database by adding more information (college, major, state, interests) in the form of a dictionary with key-value pairs. Example 2 builds a recommendation system for users to find connections with other users based on similarity.

Files

- **example1.py** – loads the extended database, stores extra attributes in a dictionary, and supports printing and searching of users across both original and added fields
- **example2.py** – implements a machine learning–based recommendation system using user features, taking an input user id and recommending other similar users
- **phase3_people.csv** – list of 10 people with attributes: id, first_name, last_name, birthdate, email, phone, college, major, state, zip, interest
- **phase3_friendships.csv** – dataset of user pairs with labels; friends = 1 means recommend connection, friends = 0 means do not recommend
- **phase3networkDB.txt** – same as the original networkDB.txt with additional information (college, major, state, interests) for each user

Compile / Run

Example 1:

- `python3 example1.py`

Example 2:

- `cd ~/Downloads/phase3`
- *(Open virtual environment)* `source venv/bin/activate`
- *(Run)* `python example2.py`

Example 1: Enlarged Database

Example 1 shows an extension of the original database using a dictionary to support dynamic attributes. It allows for flexible storage and extensive searching of user information. On running the program, user has 3 options:

1. Print all users
2. Search
3. Exit

User can search based on any domain:

```
SEARCH
=====
Search anything: usc
Louis-Dreyfus, Julia Scarlett Elizabeth
1/13/1961
(Gmail) something@email.com
(Office) 310-291-7482
College: USC
Major: Theatre
State: CA
Interest: comedy, acting, politics
-----
```

```
SEARCH
=====
Search anything: yahoo
Van Nostrand, Martin
07/24/1949
(Yahoo) myemail@yahoo.com
(Home) 2244325809
College: UCLA
Major: English
State: CA
Interest: reading, movies, hiking
-----
```

```
SEARCH
=====
Search anything: engineering
Burbank, Truman
1/17/1962
(Official) document@info.edu
(Work) 2135179989
College: Harvard
Major: Engineering
State: MA
Interest: technology, music, animals
-----
```

Issues faced:

1. When reading the information of each person, an error occurred as splitting the file by the separator creates extra empty blocks, so some lines lists are empty. Accessing elements like `lines[0]` caused an index out of range error. I had to add a check 'if `len(lines) < 5`: Continue' to skip these invalid blocks so only valid blocks of real data are processed.
2. Searching was case sensitive before normalizing to lowercase. Search is now case-insensitive.
3. Missing `.strip()` led to mismatches during storage/search due to whitespace issues.

Example 2: Recommendation System

I started by creating my own database of people with features (birthdate, email, phone number, college, major, state, zip code, and interests). The program converts the user data into feature vectors (age difference, same college, same major, same state, same zip, same email domain, shared interests). The target label is friends, where 1 means the pair should be recommended as a connection and 0 means they should not. A logistic regression model is trained using a 90% /10% train-test split and evaluated using accuracy.

After training, the user can enter a person ID, and the program recommends possible connections based on the trained model.

Issues faced:

1. I was unable to install the python packages pandas and scikit-learn in my linux terminal ('pip3 install pandas scikit-learn' fails). This is because Ubuntu blocks installing Python packages globally in the system environment, so I had to use a virtual environment (venv) instead to install locally, which isolated my project's python packages and prevented breaking system dependencies.
2. .iloc[0] will cause a crash if a user ID is not found / invalid input user ID.

Test case 1:

```
(venv) jamie@jamie-Apple-Virtualization-Generic-Platform:~/Downloads/phase3$ python example2.py

TRAINING DATA
=====
   age_difference  same_area_code  ...  shared_interests  friends
0                1                0  ...                2         1
1               39                0  ...                1         1
2               51                0  ...                2         1
3                0                0  ...                2         1
4                2                0  ...                3         1
5                2                0  ...                2         1
6                0                0  ...                2         1
7               51                0  ...                2         1
8                2                0  ...                1         1
9               11                0  ...                0          0
10              40                1  ...                0          0
11              12                0  ...                0          0
12              36                0  ...                0          0
13                0                0  ...                0          0
14                3                0  ...                0          0
15                1                0  ...                0          0
16                3                0  ...                1          0
17                2                0  ...                0          0
18              49                0  ...                1          0
19              39                1  ...                0          0

[20 rows x 9 columns]

MODEL EVALUATION
=====
Accuracy: 0.5

RECOMMEND CONNECTIONS
=====
Enter user id: 4
Anne William Berton - recommendation score: 0.98
Nayana Gupta - recommendation score: 0.91
Adam Chandler - recommendation score: 0.58
Kurt Cobain - recommendation score: 0.57
Karan Kapoor - recommendation score: 0.5
(venv) jamie@jamie-Apple-Virtualization-Generic-Platform:~/Downloads/phase3$
```

Enter user id: 4 is Amy. The model recommends similar users like Nayana or Anne because they share the same college (USC), major (Computer Engineering), state (CA) and interests (ai, coding).

Test case 2:

RECOMMEND CONNECTIONS

=====

Enter user id: 7

Katy Bruin - recommendation score: 0.85

Martin Van Nostrand - recommendation score: 0.81

Adam Chandler - recommendation score: 0.58

Kurt Cobain - recommendation score: 0.57

Anne William Berton - recommendation score: 0.53

Amy Chang - recommendation score: 0.5

Enter user id: 7 is Karan. The model recommends similar users like Katy and Martin who share similar major, state and interests.

Test case 3:

RECOMMEND CONNECTIONS

=====

Enter user id: 1

Truman Burbank - recommendation score: 0.91

Adam Chandler - recommendation score: 0.52

Enter user id: 1 is Julia. The model recommends similar users like Truman and Anne who share the same college, state and similar interests.